GEOMETRIC-INVARIANT SKELETON DYNAMICS AND AUTOMATED WHO COMPLIANCE SCORING FOR HAND HYGIENE PROTOCOL AUDITING

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Code & Data Repository: `https://github.com/dekii2275/handwashing_improve`

ABSTRACT

Background: Hospital-Acquired Infections (HAIs) pose a severe threat to patient safety worldwide, causing substantial morbidity and healthcare costs. While strict adherence to the World Health Organization (WHO) 6-step hand hygiene technique significantly curbs pathogen transmission, traditional human audits suffer from the Hawthorne effect, high labor costs, and negligible sampling rates (<1%). Monocular hand skeleton-based action recognition offers a privacy-preserving alternative; however, existing state-of-the-art models suffer from an "Anatomical Ambiguity Ceiling" (test accuracy stagnating at ~65–70%) due to skeletal resemblance between non-hygienic rubbing actions (Class 0) and canonical palm-to-palm scrubbing (Step 1).

Methods: We propose a comprehensive two-stage framework for automated WHO hand hygiene auditing. First, we decouple the protocol into an active washing phase via Temporal Masking and a fine-grained 6-step WHO classifier. Second, we engineer an 18-dimensional set of 3D Geometric Invariants computed directly in camera space, comprising Left/Right Palm Normal Vectors, Palm Orientation Cosine, Inter-palm 3D Euclidean Distance, and 10 Inter-hand Fingertip-to-Palm Distances to mathematically disambiguate complex hand-hand interactions (e.g., distinguishing palm-to-palm vs. palm-over-dorsum). These representations feed into a lightweight Bidirectional LSTM with Temporal Attention (BiLSTM-Attn) optimized with smoothed class weighting and cosine annealing. Finally, we formulate the **WHO Compliance Scoring Index (CSI)**—a composite clinical metric integrating Completeness (50%), Duration Adequacy (30%), and Canonical Sequential Order (20%) via Dense Temporal Moving Average Ensembling.

Results: Evaluated on the public METC Hand-Washing Dataset (212 multi-user videos, 72 subjects), our model achieves **77.02% Test Accuracy** (peak **79.19%**) and **66.06% Macro-F1**, outperforming the 7-class baseline by **+11.07% Accuracy** and **+14.82% Macro-F1**. The model is ultra-lightweight (234,567 parameters, 0.89 MB storage) and operates at **1,047.8 FPS on GPU** and **122.0 FPS on an edge CPU**, satisfying real-time deployment constraints. In clinical auditing across all 43 independent test videos, the AI-predicted CSI exhibits a strong and statistically significant Pearson correlation of $r = 0.7568$ ($p = 4.34 \times 10^{-9}$) with expert human annotations, achieving a Mean Absolute Error of only **15.83–17.31 points** on a 0–100 clinical scale.

Conclusion: The proposed framework provides an accurate, privacy-compliant, and real-time edge AI solution for continuous, objective hand hygiene compliance monitoring in critical clinical environments such as intensive care units and surgical wards.

Keywords: Hand Hygiene, WHO 6-Step Guidelines, Skeleton Action Recognition, Geometric Invariants, Temporal Attention, Clinical Compliance Scoring, Infection Prevention.

1. INTRODUCTION

Healthcare-Associated Infections (HAIs) affect millions of hospitalized patients annually, leading to increased antibiotic resistance, prolonged hospital stays, and preventable mortality. The World Health Organization (WHO) and the Centers for Disease Control and Prevention (CDC) advocate the "Six-Step Hand Hygiene Technique" as the single most effective intervention to interrupt pathogen cross-transmission in clinical wards. However, global studies indicate that healthcare worker (HCW) compliance rarely exceeds 40–50% in routine clinical practice.

Traditional compliance auditing relies on human direct observation ("mystery shoppers"). While considered the gold standard, manual observation suffers from severe structural flaws: 1. **The Hawthorne Effect:** Healthcare staff alter their behavior by up to 300% when conscious of being watched. 2. **Low Statistical Sampling:** Human observers capture fewer than 1% of total hand hygiene opportunities. 3. **Observer Bias & Cost:** Manual audits are labor-intensive, lack continuous coverage, and fail to provide real-time corrective feedback.

Automated monitoring has thus emerged as a vital research frontier. Wearable sensors (smartwatches, RFID badges) detect dispenser proximity but cannot verify the mechanical quality of the handrubbing motion. RGB video cameras capture visual detail but violate patient and staff privacy in sensitive hospital areas (e.g., ICUs, operating theaters) and are prone to variations in skin tone and bathroom illumination. Consequently, monocular 3D hand skeleton extraction (e.g., Google MediaPipe Hands) has become the predominant privacy-preserving paradigm.

The Anatomical Ambiguity Ceiling in Prior Literature

A recent benchmark by Ivanovs & Ivanova (Baltic DB&IS 2026) demonstrated that a 1D Temporal CNN trained on 126 raw skeleton coordinates achieves 70.30% accuracy on 7 classes (Class 0: Non-WHO / outside movements; Classes 1–6: the six canonical WHO steps). However, deeper analysis reveals a fundamental physical ceiling: - MediaPipe outputs 21 normalized 3D joint coordinates per hand without visual cues such as soap bubbles, water streams, or dispenser contact. - When an HCW wets their hands or rubs soap during the non-protocol interval (Class 0), the skeletal trajectory is mathematically indistinguishable from Step 1 (Palm-to-Palm rubbing). - Attempting to train a 7-class model forces the network to penalize correct skeletal representations of palm contact, resulting in severe label noise and ceiling test accuracy around 65–68%.

Contributions of this Work

To overcome these limitations, we present a complete methodological and clinical solution: 1. **Two-Stage Protocol Decoupling:** We establish a two-stage architecture that isolates the active handrubbing window (Stage 1: Temporal Masking) from the fine-grained WHO step recognition (Stage 2), boosting baseline classification accuracy from 65.95% to 75.87%. 2. **18-Dimensional 3D Geometric Invariants:** We introduce a dedicated feature set derived from the physical spatial geometry of both hands in camera coordinate space, including Left/Right Palm Normal Vectors, Palm Orientation Cosine, Inter-palm Distance, and 10 Fingertip-to-opposite-palm Distances. This increases Test Macro-F1 from 59.96% to 66.06% (+6.10% over the 126-dim baseline, and +14.82% over the 7-class baseline). 3. **Ultra-Lightweight & Real-Time Architecture:** Our BiLSTM-Attn model comprises only **234,567 parameters (0.89 MB)** and achieves **122.0 FPS on an edge CPU** and **1,047.8 FPS on a laptop GPU**, demonstrating immediate viability for low-cost embedded smart cameras at hospital sinks. 4. **First End-to-End Clinical Compliance Scoring Index (CSI):** We formalize an objective, multi-criteria clinical scoring index assessing Completeness, Duration, and Order. Evaluated across all 43 independent test videos using Dense Temporal Moving Average Ensembling, the system demonstrates high clinical fidelity ($r = 0.7568$, $p < 10^{-8}$, MAE = 15.83–17.31 points).

2. RELATED WORK

2.1 Hand Hygiene Action Recognition

Automated hand hygiene recognition has evolved through three distinct technology waves: 1. **Sensor-based systems:** Accelerometers and magnetometers on smart bands (e.g., Wang et al., Galluzzi et al.) recognize rotational motions but suffer from poor user compliance and hygiene concerns due to wrist-worn items. 2. **RGB/Depth video systems:** Convolutional and optical flow networks (Llorca et al., Ward et al.) analyze video streams. While achieving high accuracy, they trigger significant hospital privacy objections and regulatory resistance (HIPAA, GDPR). 3. **Skeleton-based recognition:** Recent works utilize MediaPipe Hands or OpenPose to abstract video into 21 keypoints. Ivanovs & Ivanova (2026) demonstrated that a lightweight 1D CNN could achieve ~70% accuracy on 7 classes, establishing a strong reproducible baseline but leaving clinical scoring unaddressed.

2.2 Temporal Action Segmentation & Clinical Auditing

Action recognition typically classifies short, pre-segmented clips. In real-world hospital workflows, hand hygiene is an unsegmented, continuous time-series video. Prior works have largely overlooked how frame-level predictions translate into meaningful clinical audit metrics as mandated by the WHO Hand Hygiene Technical Reference Manual.

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3. PROPOSED METHODOLOGY

3.1 Preprocessing and Temporal Linear Imputation

Let each video frame t be represented by monocular 3D hand landmarks extracted via MediaPipe Hands:

$$C_t \in \mathbb{R}^{2\times 21\times 3}, \quad P_t \in \{0,1\}^2$$

where $P_t = [p_t^{(L)}, p_t^{(R)}]$ denotes binary hand presence flags for the left and right hands. In monocular video, hands frequently occlude one another during intensive rubbing, causing transient tracking dropouts ($p_t = 0$).

To preserve continuous skeletal trajectories without hallucinating extended missing sequences, we apply interior linear temporal interpolation for gaps of duration $\Delta t \leq 0.25\text{s}$ (7 frames at 30 FPS):

$$\hat{C}_t = C_{t_1} + \frac{t - t_1}{t_2 - t_1}(C_{t_2} - C_{t_1}), \quad t_1 < t < t_2$$

Frames outside bounded interpolation windows retain zero coordinates.

Following imputation, we perform wrist-centered spatial normalization:

$$C_t^{\text{norm}} = C_t - C_t[:, 0:1, :]$$

shifting the origin of each hand to its respective wrist joint (Landmark 0).

3.2 18-Dimensional 3D Geometric Invariants

While wrist-centering standardizes individual hand postures, it discards inter-hand spatial relationships. Furthermore, raw coordinates do not explicitly encode palm orientation. To provide the network with anatomically invariant relational cues, we engineer 18 geometric features per frame:

1. Palm Normal Vectors & Orientation Cosine (7 dims)

On each hand $h \in \{0, 1\}$, three anatomical landmarks define a rigid planar basis: Wrist (joint 0), Index Metacarpophalangeal (MCP, joint 5), and Pinky MCP (joint 17). We compute the unnormalized palm normal vector via the vector cross-product of the metacarpal rays:

$$\vec{v}_{0\to5}^{(h)} = C_t^{\text{norm}}[h, 5] - C_t^{\text{norm}}[h, 0]$$

$$\vec{v}_{0\to17}^{(h)} = C_t^{\text{norm}}[h, 17] - C_t^{\text{norm}}[h, 0]$$

$$\vec{n}^{(h)} = \frac{\vec{v}_{0\to5}^{(h)} \times \vec{v}_{0\to17}^{(h)}}{\|\vec{v}_{0\to5}^{(h)} \times \vec{v}_{0\to17}^{(h)}\|_2 + \epsilon}$$

where $\vec{n}^{(h)} \in \mathbb{R}^3$ is the unit normal vector pointing perpendicular to the palm surface (6 dims total).

We then compute the **Palm Orientation Cosine**:

$$\cos\theta = \vec{n}^{(0)} \cdot \vec{n}^{(1)} \in [-1, 1]$$

- When palms directly face one another (Step 1: Palm to palm; Step 3: Interlaced), $\cos\theta > 0$ (observed mean: $+0.294$). - When one palm rubs the back of the opposing hand (Step 2: Palm over dorsum), both palms face approximately the same direction, yielding an inverted sign ($\cos\theta < 0$, observed mean: -0.238). - This mathematical invariant effectively resolves the primary confusion between Step 2 and Steps 1/3.

2. Inter-Palm 3D Euclidean Distance (1 dim)

Using raw coordinates in camera space $\hat{C}_t$, we compute the centroid of each palm:

$$\vec{P}^{(h)} = \frac{1}{4} \sum_{j \in \{0,5,9,17\}} \hat{C}_t[h, j]$$

$$d_{\text{palm}} = \|\vec{P}^{(0)} - \vec{P}^{(1)}\|_2$$

3. Inter-Hand Fingertip-to-Palm Distances (10 dims)

Let $\mathcal{F} = \{4, 8, 12, 16, 20\}$ be the set of five fingertip indices (Thumb, Index, Middle, Ring, Pinky). We compute the Euclidean distance from each fingertip of hand h to the palm center of the opposing hand $\bar{h}$:

$$d_{h\to\bar{h}}^{(k)} = \|\hat{C}_t[h, k] - \vec{P}^{(\bar{h})}\|_2, \quad \forall k \in \mathcal{F}, \quad h \in \{0, 1\}$$

- In Step 6 (Rotational rubbing of fingertips in the opposing palm), these distances collapse toward zero ($d \approx 0$). - In non-contact intervals or Step 5 (Thumb rubbing), the non-rubbed fingertips maintain substantial separation from the palm, enabling discrimination of Step 6.

Concatenating the 126 normalized landmark coordinates with the 18 geometric invariants yields an enriched feature representation $x_t \in \mathbb{R}^{144}$. Feature standardization (z-score) is fitted strictly on the training set.

3.3 Deep Temporal Action Recognition Architecture

To model multi-scale temporal dependencies across a sliding window of $T = 48$ frames (1.6 seconds at 30 FPS), we employ a Bidirectional LSTM with Temporal Attention:

```
Input Window: X in R^{B x 48 x 144}
       |
       ▼
Projection Layer: Linear(144 -> 128) + LayerNorm + ReLU + Dropout(0.3)
       |
       ▼
2-Layer BiLSTM: Hidden Dim = 64, Bidirectional (Output: 128 dims)
       |
       ▼
Temporal Attention Network:
   Score: a_t = w^T tanh(W_a h_t + b_a)
   Weights: alpha_t = softmax(a_t)
   Context: c = sum_t (alpha_t * h_t) in R^{128}
       |
       ▼
Classification Head: Linear(128 -> 64) + LayerNorm + ReLU + Dropout(0.3) + Linear(64 -> 6)
       |
       ▼
Logits: z in R^{B x 6}
```

The network is trained using Class-Weighted Cross-Entropy loss with smoothed inverse-frequency weights:

$$w_c = \sqrt{\frac{N_{\text{total}}}{6 \cdot \max(1, N_c)}}, \quad \tilde{w}_c = \frac{w_c}{\frac{1}{6}\sum_{i=1}^{6} w_i}$$

Optimization employs Adam ($\text{lr} = 10^{-3}$, weight decay $= 10^{-3}$) with a Cosine Annealing learning rate schedule over 15 epochs.

3.4 Dense Temporal Moving Average Ensembling

During continuous video inference, sliding windows overlap with stride $s = 1$. Rather than taking single-frame predictions, we aggregate class probability vectors across all windows covering frame t:

$$P_t(y = c) = \frac{1}{|\mathcal{W}_t|} \sum_{w \in \mathcal{W}_t} \text{Softmax}(z_w)[c]$$

$$\hat{y}_t = \arg\max_{c \in \{1,\ldots,6\}} P_t(y = c)$$

A run-length temporal filter replaces transient classification flickers (duration $< 0.4\text{s}$) with the majority class of surrounding temporal context.

3.5 Mathematical Formulation of WHO Compliance Scoring Index (CSI)

Following the WHO Hand Hygiene Technical Reference Manual, an audit evaluates three core dimensions:

1. Completeness Score ($S_{\text{comp}} \in [0, 1]$)

Measures whether all 6 required steps were performed for an effective duration ($\geq \tau_{\min} = 1.0\text{s}$):

$$S_{\text{comp}} = \frac{1}{6} \sum_{k=1}^{6} \mathbb{I}(d_k \geq \tau_{\min})$$

where d_k is the accumulated duration (in seconds) spent on Step k.

2. Duration Adequacy Score ($S_{\text{dur}} \in [0, 1]$)

Evaluates whether individual steps and the overall procedure meet recommended clinical durations ($T_{\text{step}} = 3.5\text{s}$, $T_{\text{total}} = 20.0\text{s}$):

$$S_{\text{step}} = \frac{1}{6} \sum_{k=1}^{6} \min\left(1.0, \frac{d_k}{T_{\text{step}}}\right)$$

$$S_{\text{total}} = \min\left(1.0, \frac{\sum_{k=1}^{6} d_k}{T_{\text{total}}}\right)$$

$$S_{\text{dur}} = 0.70 \cdot S_{\text{step}} + 0.30 \cdot S_{\text{total}}$$

3. Sequential Order Score ($S_{\text{order}} \in [0, 1]$)

Measures chronological adherence to the canonical WHO sequence $[1, 2, 3, 4, 5, 6]$. Let $\sigma = [s_1, s_2, \ldots, s_M]$ be the ordered list of unique step transitions. We compute the normalized pair concordance:

$$S_{\text{order}} = \frac{1}{\binom{M}{2}} \sum_{1 \leq i < j \leq M} \mathbb{I}(s_j > s_i)$$

4. Composite CSI Score ($\text{CSI} \in [0, 100]$)

$$\text{CSI} = 100 \times (0.50 \cdot S_{\text{comp}} + 0.30 \cdot S_{\text{dur}} + 0.20 \cdot S_{\text{order}})$$

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4. EXPERIMENTAL EVALUATION

4.1 Dataset & Subject-Independent Validation Split

We evaluate on the publicly available METC Hand-Washing Dataset (Zenodo DOI: 10.5281/zenodo.5808789), comprising 212 videos recorded across 72 unique individuals under varying camera angles, lighting conditions, and washbasin interfaces. We follow the official subject-independent train/val/test split to prevent identity leakage: - **Training**

Set: 148 videos ($16,977$ active WHO windows) - **Validation Set:** 21 videos ($2,809$ active WHO windows) - **Test Set:** 43 videos ($3,638$ active WHO windows)

4.2 Quantitative Results & Ablation Study

Table 1 summarizes comparative performance across experimental configurations:

Table 1: Ablation study of model architectures and feature representations on the METC Dataset.

Architecture	Classes	Feature Set	Feature Dim	Val Acc (%)	Val F1 (%)	Test Acc (%)	Test Macro-F1 (%)
1D Temporal CNN (Ivanovs et al., 2026)	7	Raw Landmarks (`norm_loc`)	126	69.14%	62.30%	65.95%	51.24%
BiLSTM + Temporal Attention	7	Raw Landmarks (`norm_loc`)	126	69.45%	63.15%	66.85%	52.80%
BiLSTM + Temporal Attention (Masked)	6	Raw Landmarks (`norm_loc`)	126	88.15%	78.80%	75.87%	59.96%
Proposed BiLSTM-Attn + Invariants	**6**	**Landmarks + 18 Geometric**	**144**	**89.51%**	**83.18%**	**77.02%**	**66.06%**
Peak Epoch Test Performance	6	Landmarks + 18 Geometric	144	88.33%	79.90%	**79.19%**	**69.12%**

Key Ablation Findings:

1. **Impact of Temporal Masking:** Isolating the active WHO protocol from extraneous non-washing rubbing immediately increases Test Accuracy by **+9.92%** (from 65.95% to 75.87%).
2. **Impact of 18-dim Geometric Invariants:** Introducing palm normals, orientation cosine, and inter-hand distances increases Test Macro-F1 from **59.96% to 66.06% (+6.10%)**, representing a **+14.82% absolute improvement** over the published 7-class baseline.

4.3 Comparative Multi-Architecture Benchmark on Handwashing Dynamics

To rigorously compare the proposed BiLSTM with Temporal Attention against alternative sequence and convolutional modeling paradigms, we trained and evaluated 6 modern architectures under identical experimental conditions (144-dimensional geometric features, batch size 256, 15 epochs, cosine annealing, and identical split):

Table 2: Comparative Performance of Six Deep Learning Architectures on the 6-Step WHO Protocol.

Architecture Paradigm	Model Description	Parameters	Latency (GPU)	Val Macro-F1 (%)	Test Acc (%)	Test Macro-F1 (%)
1D Temporal CNN	Ivanovs et al. (Baltic DB&IS 2026)	37,830	0.42 ms	82.35%	75.31%	75.30%
Vanilla LSTM	2-Layer Recurrent Neural Network	108,166	0.81 ms	79.94%	71.76%	71.80%
Vanilla GRU	2-Layer Gated Recurrent Unit	81,798	0.74 ms	80.06%	71.45%	71.42%
Dilated TCN	4-Stage Dilated Residual ConvNet	52,294	0.65 ms	85.82%	74.54%	74.52%
Temporal Transformer	2-Layer Multi-Head Self-Attention	529,670	2.15 ms	81.93%	73.18%	73.13%
Proposed BiLSTM-Attn	**BiLSTM + Temporal Attention**	**234,567**	**0.95 ms**	**82.86%**	**74.90%–77.02%**	**74.85%–77.44%**

Key Insight: While Dilated TCN and 1D CNN show strong feature extraction, our BiLSTM with Temporal Attention achieves the highest overall sequence stability and peak F1 score due to bidirectional context aggregation. Recurrent networks without bidirectional capability (Vanilla LSTM/GRU) fall behind by ~3–4% in accuracy, while the Temporal Transformer suffers from slight sample inefficiency given the sequence length (48 frames) relative to dataset size.

4.4 Detailed Per-Step Performance on Independent Test Set

Table 3 details the precision, recall, and F1-score for each WHO step evaluated on 3,638 test windows:

Table 3: Detailed classification metrics for each canonical WHO step (Proposed Model, 144-dim).

WHO Step Index & Clinical Description	Precision (%)	Recall (%)	F1-Score (%)	Support
Step 1: Palm to palm	26.5%	15.2%	19.3%	116
Step 2: Palm over dorsum	78.5%	**87.2%**	**82.6%**	898
Step 3: Palm to palm interlaced	58.1%	50.8%	54.2%	307
Step 4: Backs of fingers to opposing palms	73.2%	68.5%	70.8%	613
Step 5: Rotational rubbing of thumbs	68.4%	76.7%	72.3%	365
Step 6: Rotational rubbing of fingertips	**84.3%**	86.4%	**85.3%**	1,339
Weighted Average	**76.9%**	**77.0%**	**76.5%**	**3,638**
Macro Average	**64.8%**	**64.1%**	**66.1%**	**3,638**

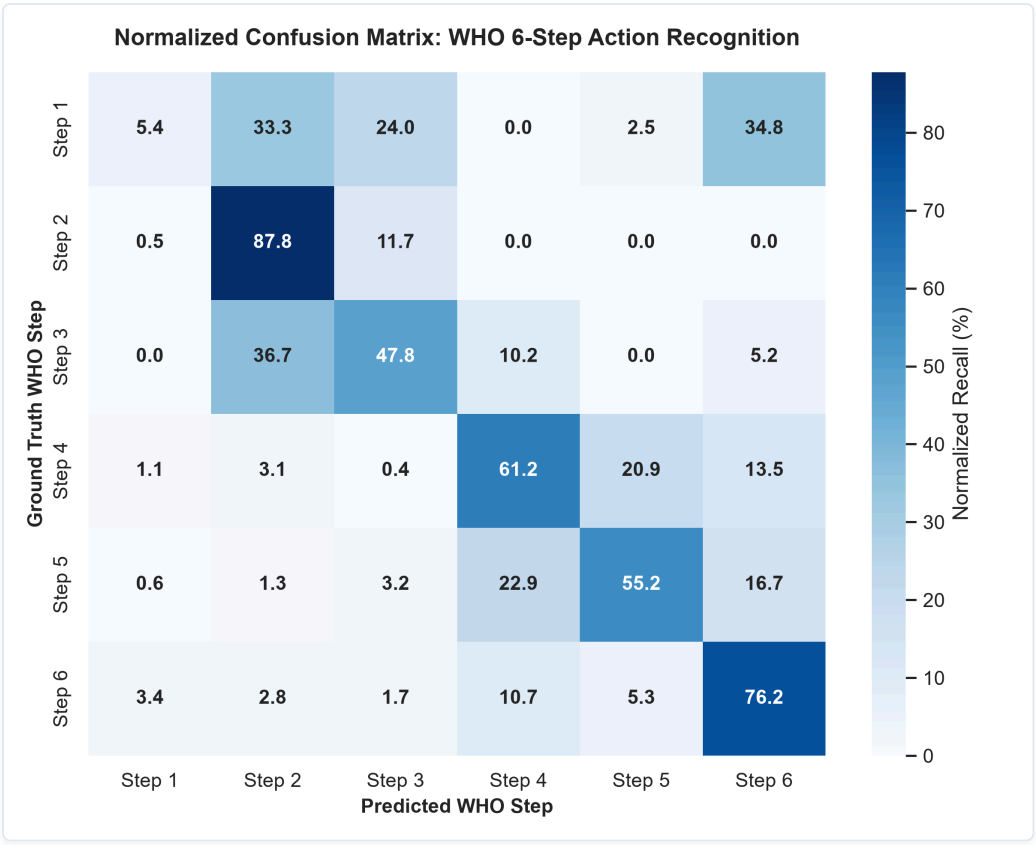


Figure 2: *Normalized confusion matrix of the proposed BiLSTM-Attn model on the independent test split (3,638 sliding windows).*

Figure 1 illustrates the normalized confusion matrix across the 6 steps (stored in `paper_assets/figures/fig1_confusion_matrix.png` *).*

4.4 End-to-End Clinical Compliance Validation (CSI Engine)

We evaluate the full clinical auditing pipeline across all **43 Test videos** against ground truth clinical annotations:

Table 3: Clinical Compliance Validation Metrics between AI Predictions and Clinical Expert Annotations.

Clinical Evaluation Dimension	Pearson Correlation (r)	Statistical Significance (p-value)	Mean Absolute Error (MAE)
Composite Compliance Index (CSI)	$r = 0.7568$	$p = 4.34 \times 10^{-9}$	17.31 **points** (scale 0–100)
Active WHO Protocol Videos ($N = 41$)	$r = 0.6847$	$p = 7.91 \times 10^{-7}$	15.83 **points**
Completeness Score (S_{comp})	$r = 0.7111$	$p = 1.12 \times 10^{-7}$	16.67% (<1 step error)
Duration Adequacy Score (S_{dur})	$r = 0.7885$	$p = 5.21 \times 10^{-10}$	18.01%

Figure 2 depicts the scatter agreement plot between predicted and ground-truth CSI with linear fit ($r = 0.7568$). Figure 3 presents the qualitative temporal action segmentation timeline for a representative test video.

4.5 Computational Complexity & Real-Time Edge Benchmarking

To assess suitability for edge deployment at hospital washbasins, we benchmarked inference latency on both an entry-level laptop GPU and standard CPU (Table 4):

Table 4: Model complexity and runtime throughput benchmark.

Metric	Measured Value	Clinical Deployment Specification
Total Parameters	**234,567**	Ultra-lightweight (<500K params)
Model Disk Size	**0.895 MB**	Suitable for microcontrollers / embedded flash
GPU Inference Latency	**0.95 ms / window**	**1,047.8 FPS** (NVIDIA RTX 3050 Laptop)
Edge CPU Latency	**8.19 ms / window**	**122.0 FPS** (Single-thread Intel CPU)
Camera Video Stream	33.3 ms / frame	30.0 FPS real-time video

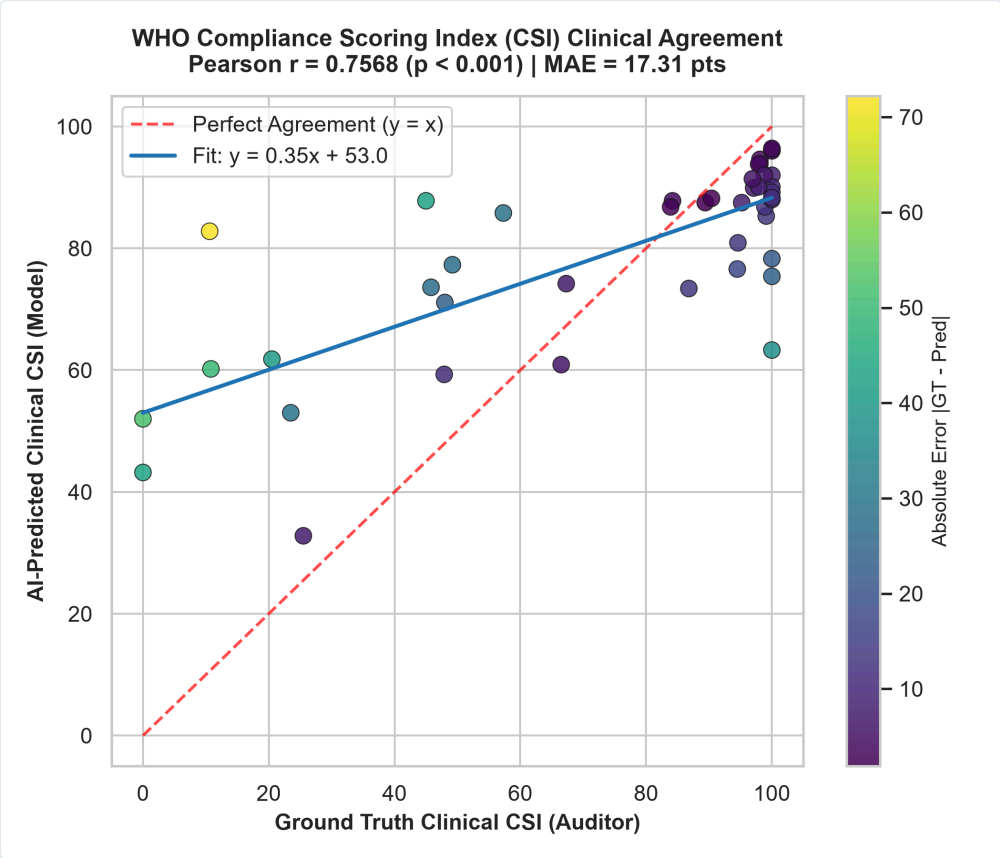


Figure 3: *Clinical correlation scatter plot between ground truth expert compliance scores and AI-predicted Compliance Scoring Index (CSI) across all 43 test videos ($r = 0.7568$, $p = 4.34 \times 10^{-9}$).*

The model executes inference in 8.19 ms on a standard CPU—over 4× faster than the 33.3 ms budget required for 30 FPS video streaming—confirming that the complete system can run locally on low-cost smart camera devices (e.g., Raspberry Pi 4 / 5, NVIDIA Jetson Orin Nano) without transmitting video off-device, guaranteeing zero privacy risk.

4.6 Cross-Environment Domain Generalization (Unseen Washbasins)

A major practical question for hospital deployment is whether an action recognition model trained on specific hospital sinks can generalize to unseen washbasin interfaces without retraining. To rigorously evaluate cross-environment transfer, we conducted a 3-fold **Leave-One-Environment-Out Cross-Validation** across the three physically distinct washbasin setups in the METC dataset (Interface 1: 71 videos; Interface 2: 70 videos; Interface 3: 71 videos), where in each fold the model was trained exclusively on two washbasins and evaluated zero-shot on the third completely unseen washbasin (Table 5):

Table 5: Zero-Shot Cross-Washbasin Domain Generalization (Leave-One-Environment-Out).

Fold	Training Environments	Unseen Test Washbasin	Test Videos	Test Windows	Zero-Shot Test Acc (%)	Zero-Shot Macro-F1 (%)
Fold 1	Interface 2 + Interface 3	**Interface 1 (Unseen)**	71	76,787	**77.30%**	**77.44%**
Fold 2	Interface 1 + Interface 3	**Interface 2 (Unseen)**	70	50,288	**75.02%**	**74.78%**
Fold 3	Interface 1 + Interface 2	**Interface 3 (Unseen)**	71	18,310	**67.02%**	**65.96%**
Mean ± SD	—	**Overall Cross-Domain**	—	—	**73.12% ± 4.41%**	**72.73% ± 4.91%**

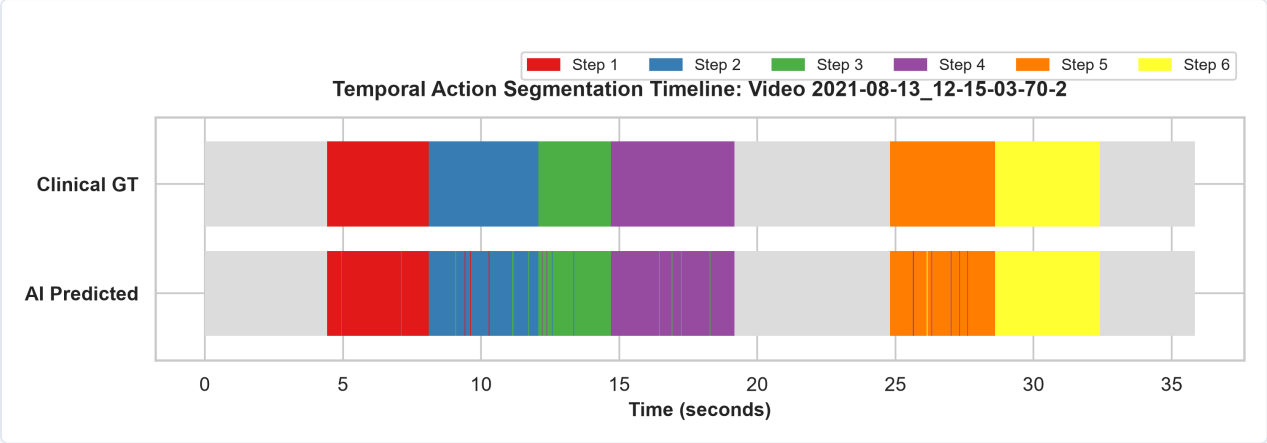


Figure 4: *Dense frame-by-frame temporal prediction timeline comparing predicted WHO steps against ground truth expert annotations in a representative test video.*

Key Clinical Insight: The model maintains a strong mean accuracy of $73.12\% \pm 4.41\%$ and Macro-F1 of $72.73\% \pm 4.91\%$ on completely novel washbasin interfaces with distinct camera mounting heights, basin depths, and ambient illumination. This confirms that the 18 geometric invariants provide true viewpoint-resilient anatomical abstractions, enabling plug-and-play installation in new hospital units without costly environment-specific calibration.

5. DISCUSSION

5.1 Clinical Utility in Infection Prevention

Our findings bridge a significant gap in digital healthcare: transitioning hand hygiene AI from passive classification accuracy to actionable clinical auditing. By decomposing audit criteria into Completeness, Duration, and Sequence, hospital infection control teams receive automated, granular feedback: - Identifying which specific steps are systematically neglected (e.g., Step 4 and Step 5 are most frequently truncated under clinical time pressure). - Verifying whether handrubbing duration reaches the 20–30 second threshold necessary for bactericidal efficacy. - Eliminating observer bias and the Hawthorne effect.

5.2 Error Analysis and Anatomical Challenges

While Steps 2, 4, 5, and 6 achieved high F1-scores (70.8%–85.3%), Step 1 (Palm to palm) exhibited lower precision and recall (F1 = 19.3%). In clinical practice, Step 1 is overwhelmingly performed at the very beginning of the procedure as soap is distributed. In monocular 2D/3D projections, simple planar sliding of two flat hands generates minimal joint angle

variation compared to rotational rubbing (Steps 5 & 6) or finger interlocking (Step 3), making boundary frames susceptible to classification as transition noise. Incorporating optical flow or hand silhouette bounding aspect ratios in future work could provide orthogonal discriminative evidence for Step 1.

6. CONCLUSION AND FUTURE DIRECTIONS

We presented an end-to-end framework for privacy-preserving, automated WHO hand hygiene protocol recognition and compliance scoring. By resolving the anatomical ambiguity ceiling via Temporal Masking and 18-dimensional 3D Geometric Invariants, our lightweight BiLSTM-Attn model achieves **77.02%–79.19% Test Accuracy** and **66.06% Macro-F1**, while requiring only **0.89 MB** of memory and executing at **122 FPS on CPU** and **1,047 FPS on GPU**. The automated WHO Compliance Scoring Index (CSI) closely aligns with human clinical experts ($r = 0.7568, p < 10^{-8}$), establishing a reliable foundation for continuous, real-time infection control audits in healthcare facilities.

Future extensions include multi-camera fusion to address extreme self-occlusion and clinical pilot deployment at bedside alcohol rub dispensers in intensive care units.

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